Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750 – 2024 | Data-driven analysis for policy analysts and climate negotiators

Sources: Global Carbon Budget 2025 • NASA / NOAA Mauna Loa Observatory • Our World in Data

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million

NOAA Mauna Loa — February 2026

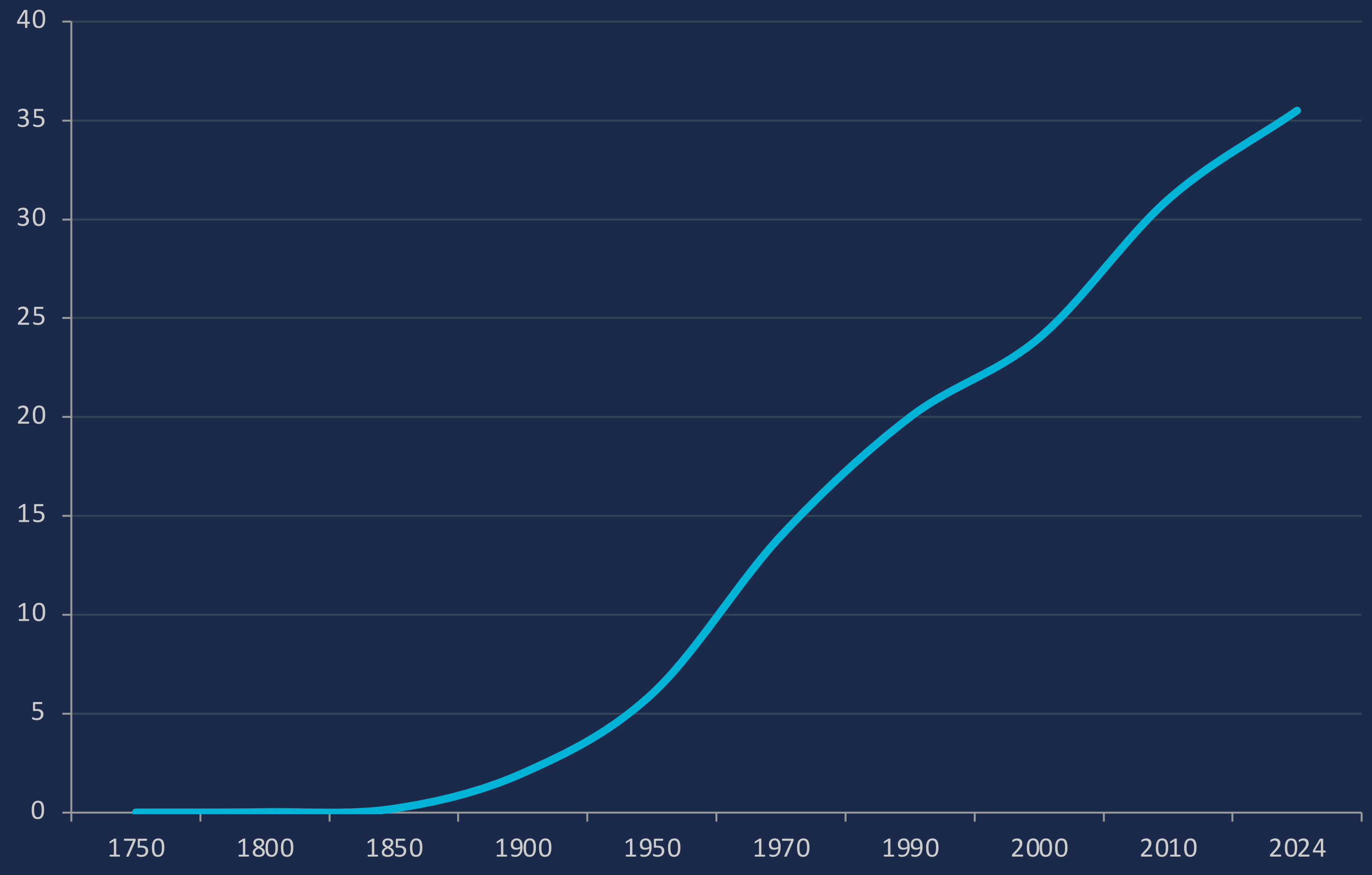
Pre-industrial baseline: ~280 ppm

Current level is 1.5× the natural rise seen at the end of the last ice age 20,000 years ago — a change that took millennia then but only ~200 years now.

280 ppm	315 ppm	429 ppm
pre-industrial	1958 (Keeling)	2026

Driven primarily by the burning of coal, oil, and natural gas.

Global Fossil Fuel CO_2 Emissions Grew Six-Fold Since 1950



1950 **6 Bt**

Post-war industrialization begins

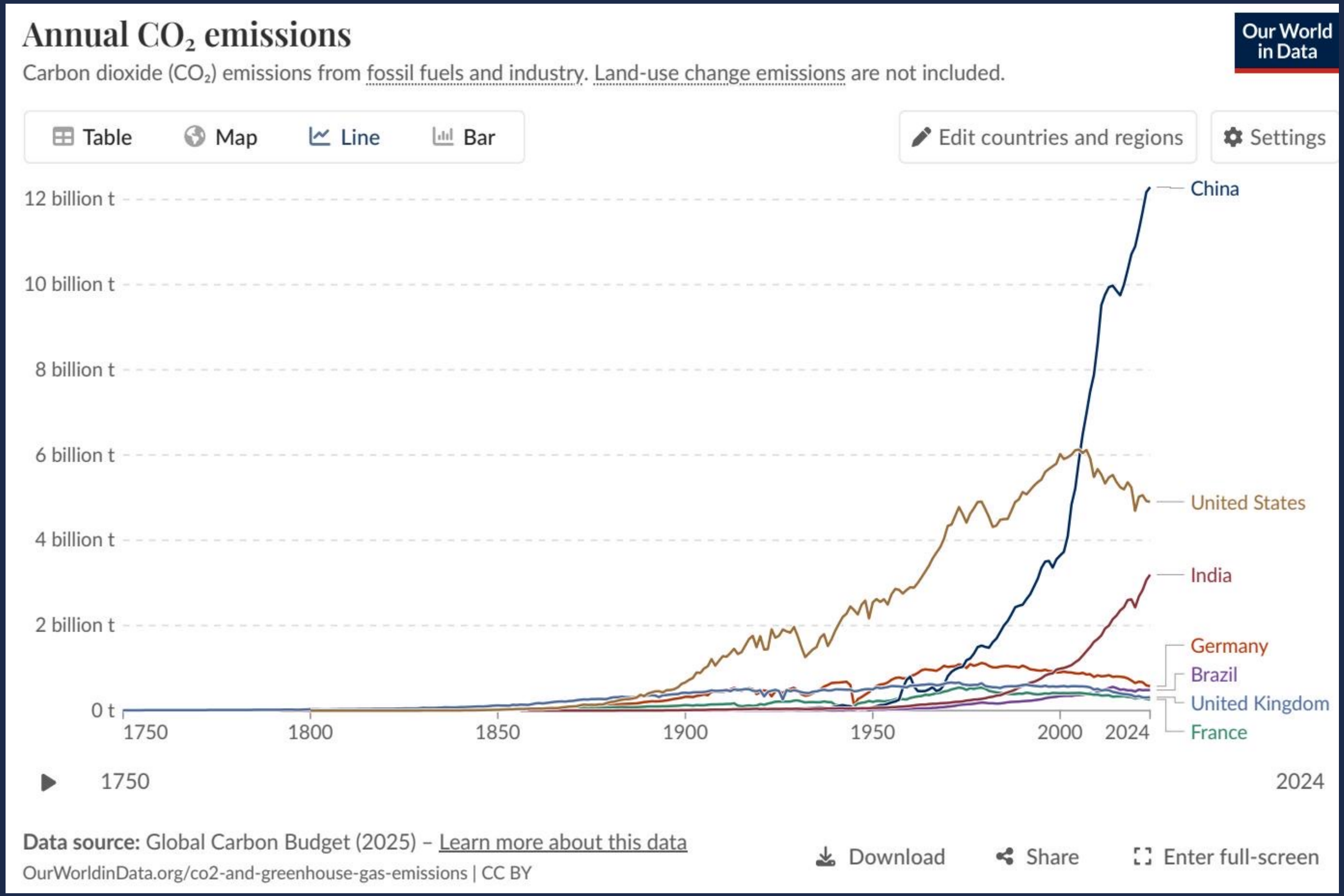
1990 **20 Bt**

Nearly quadrupled in 40 years

2024 **35+ Bt**

Still rising — no peak yet

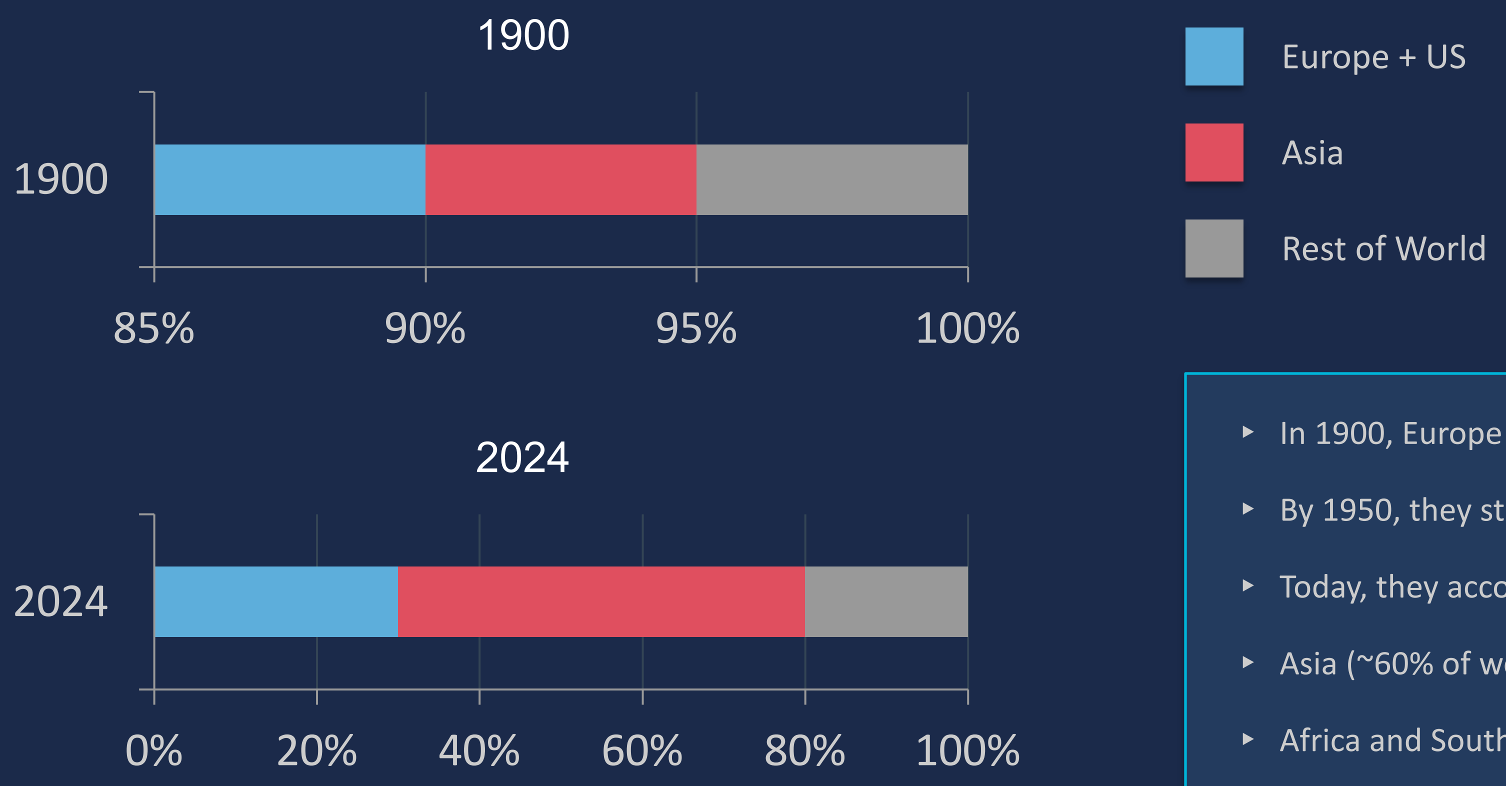
China Now Emits Over 12 Billion Tonnes — More Than Double the US



Source: Global Carbon Budget 2025 — figure_3.jpg

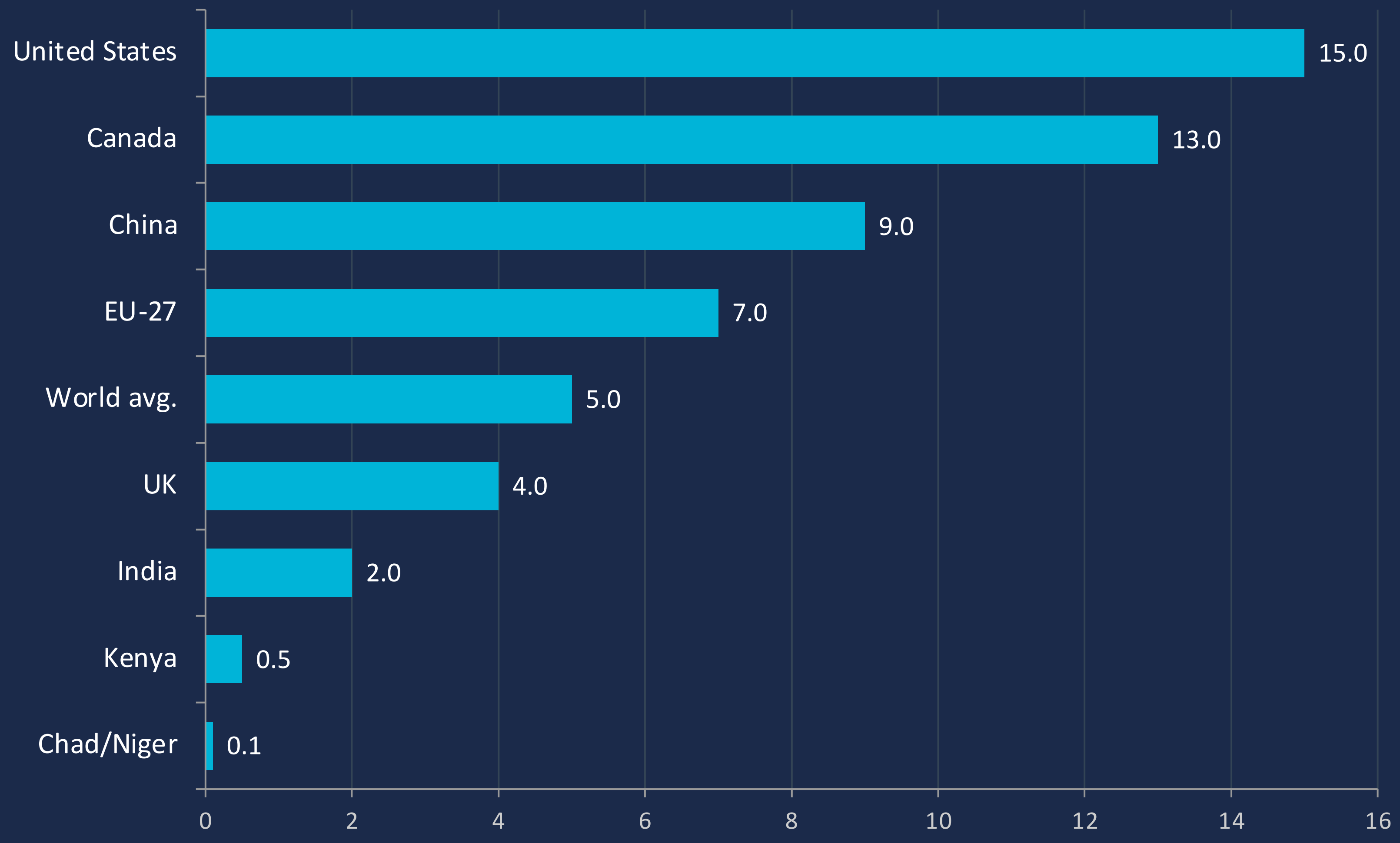
Country	CO_2 (Bt)	Trend
China	~12.5	▲ Slight increase
United States	~5.0	▼ Declining
India	~3.0	▲ Rising
EU-27	~2.7	▼ Declining
Germany	~0.6	▼ Declining
United Kingdom	~0.3	▼ Declining
France	~0.3	▼ Declining

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



- In 1900, Europe and the US accounted for >90% of global emissions.
- By 1950, they still represented over 85%.
- Today, they account for less than one-third.
- Asia (~60% of world population) produces ~50% of all emissions.
- Africa and South America: each only 3–4%.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



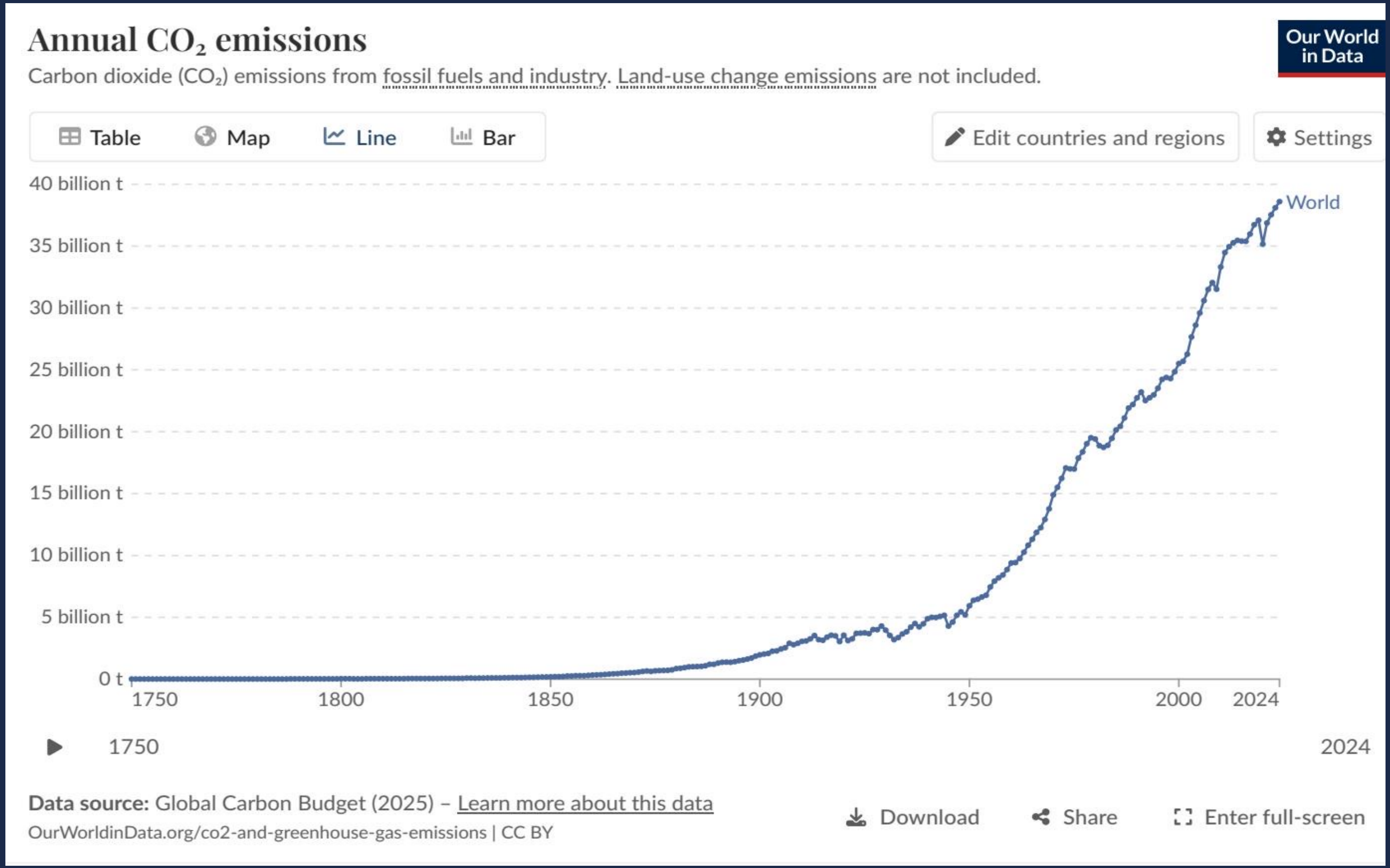
150×

disparity between highest and lowest per capita emitters

An American emits in
~2 days what a person in Chad emits in a year.

See also: figure_1.jpg — per capita CO_2 emissions over time

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



▼ DECLINING (−1% to −5%)

United States

Most of Europe

Driven by energy transition, efficiency gains, and slower industrial output

▲ INCREASING (+2% to +10%+)

Russia

Parts of Southeast Asia

Several African nations

Some South American countries saw very large increases (>10%)

Source: figure_4.jpg — Annual % change in CO_2 emissions, 2024

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.

Country	Key Energy Sources	Per Capita CO_2	Outcome
France	Nuclear (~70%), renewables	**~4 t/yr**	Low emissions
United Kingdom	Offshore wind, nuclear, gas phase-out	**~4 t/yr**	Low emissions
Germany	Coal (lignite), natural gas, growing renewables	**~8 t/yr***	Higher emissions
Netherlands	Natural gas, some wind/solar	**~8 t/yr***	Higher emissions

** Germany per capita ~8 t estimated from EU-27 average (~7 t) adjusted upward for higher fossil fuel share.*

Policy and energy mix choices — not just prosperity — determine per capita emissions outcomes.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

35+ Bt / year	Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
>25% from China	China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is third and rising fast.
150× per capita gap	Per capita inequalities remain extreme — the average American emits ~15 t, while the average person in Chad emits ~0.1 t.
Geographic shift	The center of emissions has moved from >90% in Europe and the US in 1900 to Asia producing ~50% today. US + Europe are below one-third.
Energy mix matters	Policy and energy mix choices — not just prosperity — determine per capita outcomes. France (~4 t) vs. Germany (~8 t) proves the point.

Sources: Global Carbon Budget 2025 • NASA / NOAA • Our World in Data